

Auszug aus:

Rubidium 87 D Line Data

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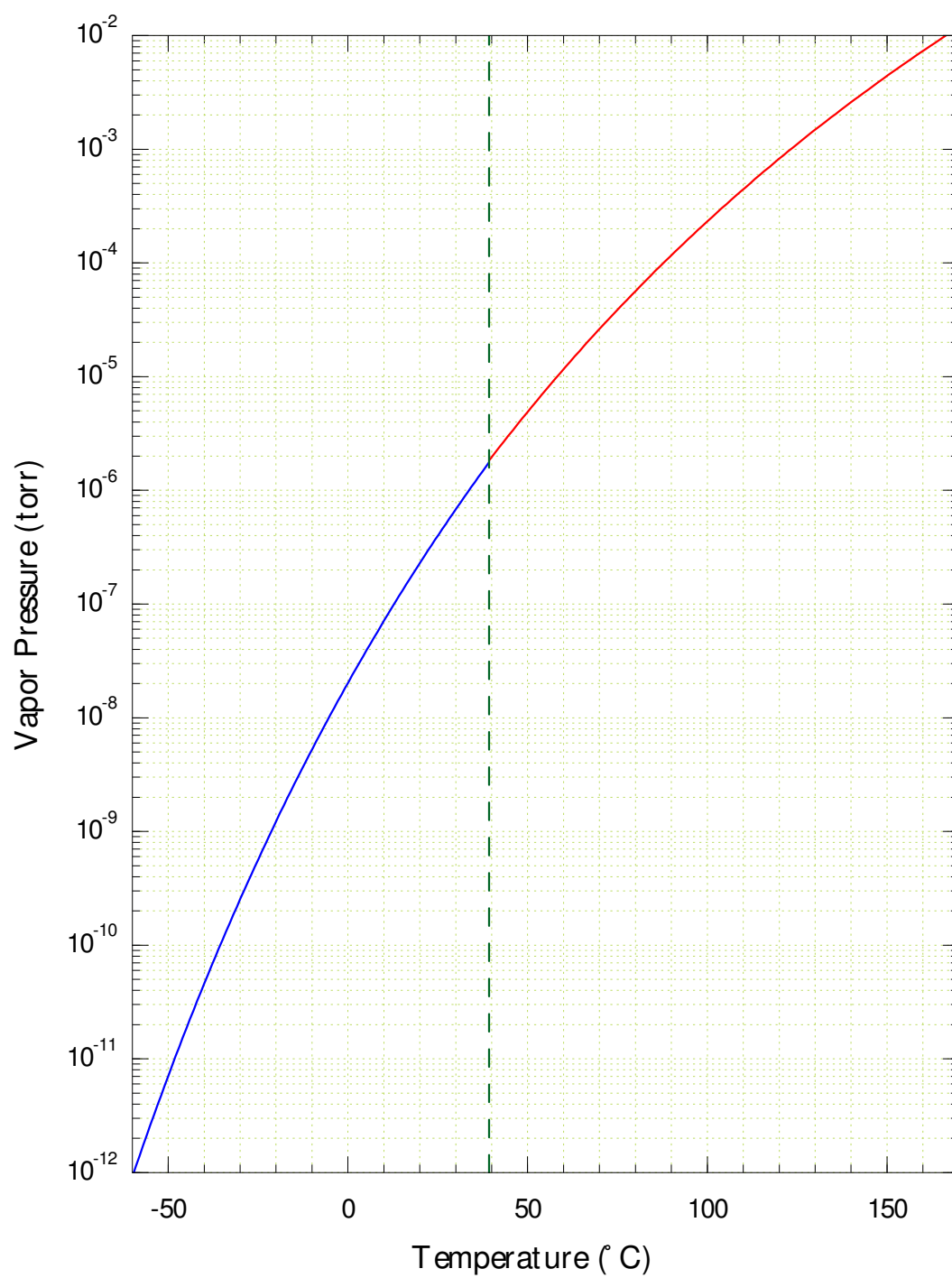


Figure 1: Vapor pressure of rubidium from the model of Eqs. (1). The vertical line indicates the melting point.

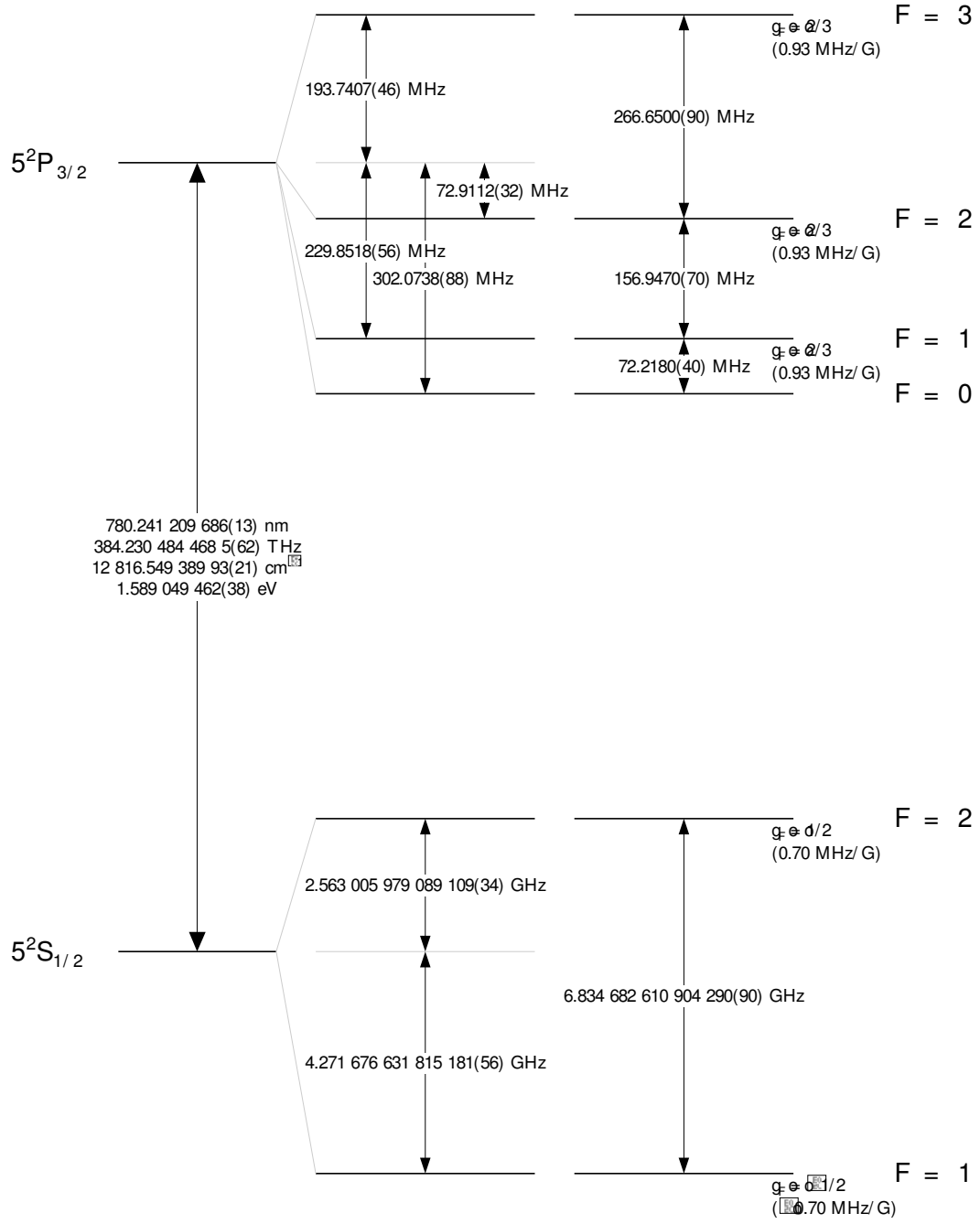


Figure 2: Rubidium 87 D_2 transition hyperfine structure, with frequency splittings between the hyperfine energy levels. The excited-state values are taken from [9], and the ground-state values are from [29]. The relative hyperfine shifts are shown to scale within each hyperfine manifold (but visual spacings should not be compared between manifolds or to the optical splitting). The approximate Landé g_F -factors for each level are also given, with the corresponding Zeeman splittings between adjacent magnetic sublevels.

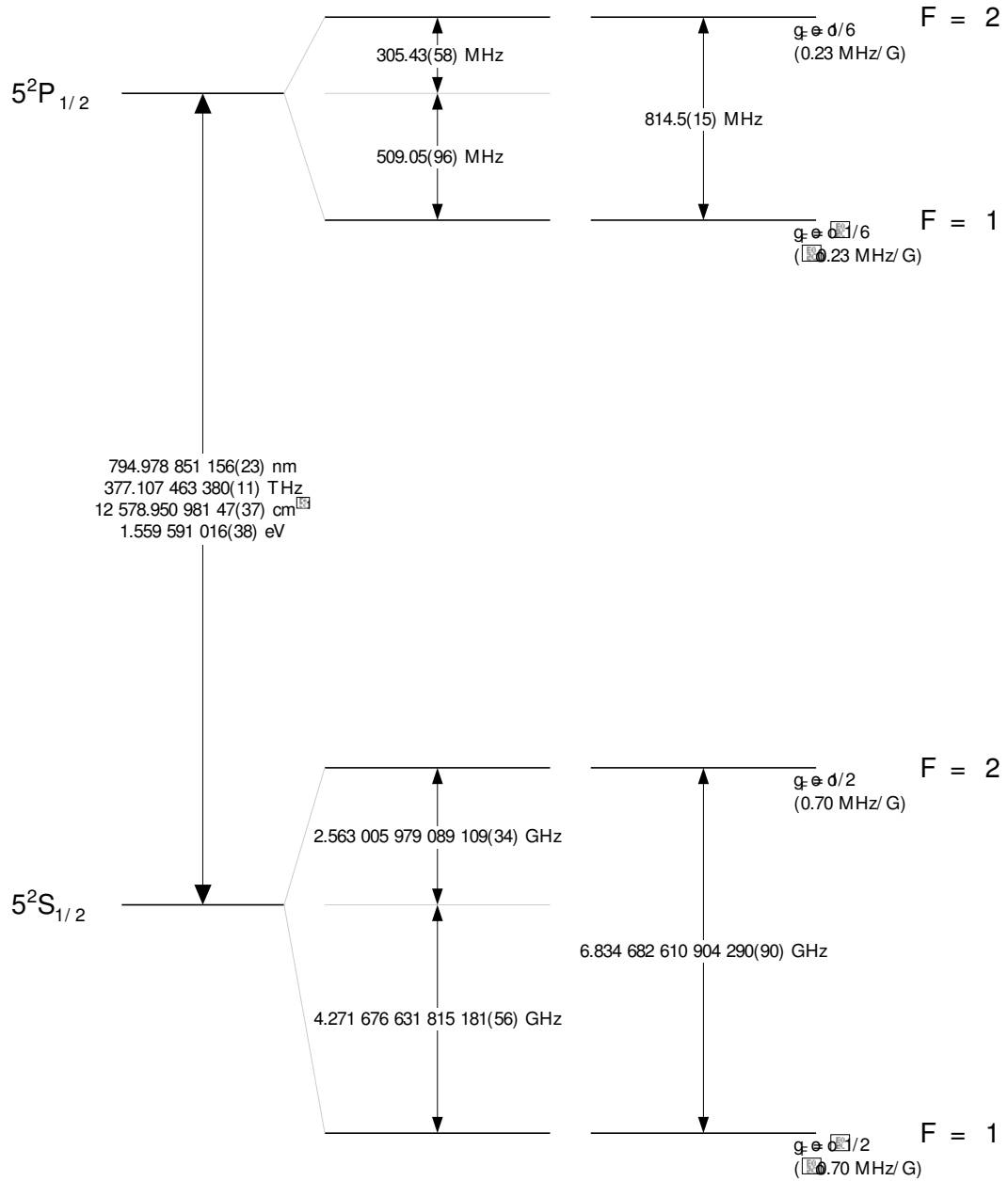


Figure 3: Rubidium 87 D_1 transition hyperfine structure, with frequency splittings between the hyperfine energy levels. The excited-state values are taken from [10, 11, 26], and the ground-state values are from [29]. The relative hyperfine shifts are shown to scale within each hyperfine manifold (but visual spacings should not be compared between manifolds or to the optical splitting). The approximate Landé g_F -factors for each level are also given, with the corresponding Zeeman splittings between adjacent magnetic sublevels.

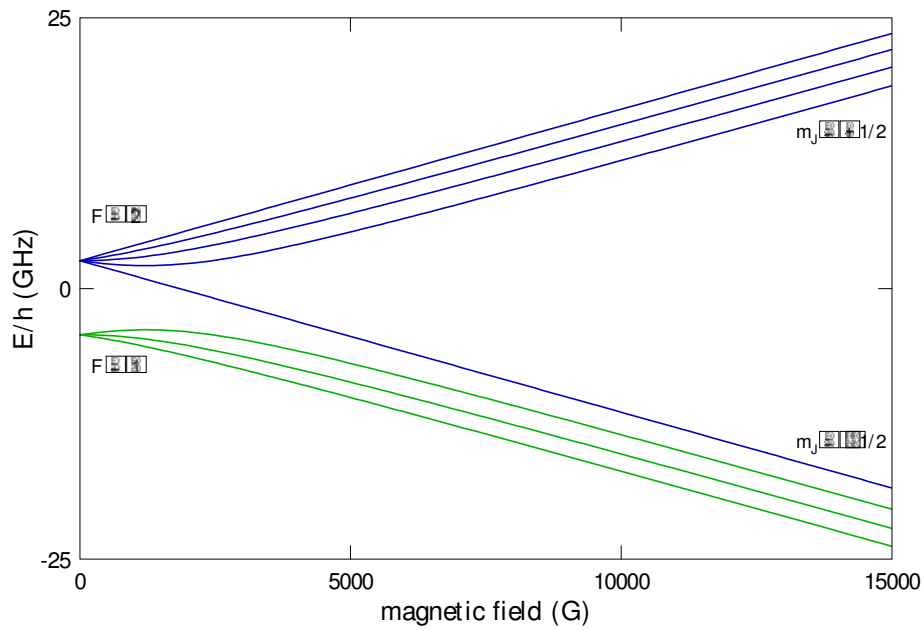


Figure 4: Rubidium 87 $5^2S_{1/2}$ (ground) level hyperfine structure in an external magnetic field. The levels are grouped according to the value of F in the low-field (Zeeman) regime and m_j in the strong-field (hyperfine Paschen-Back) regime.

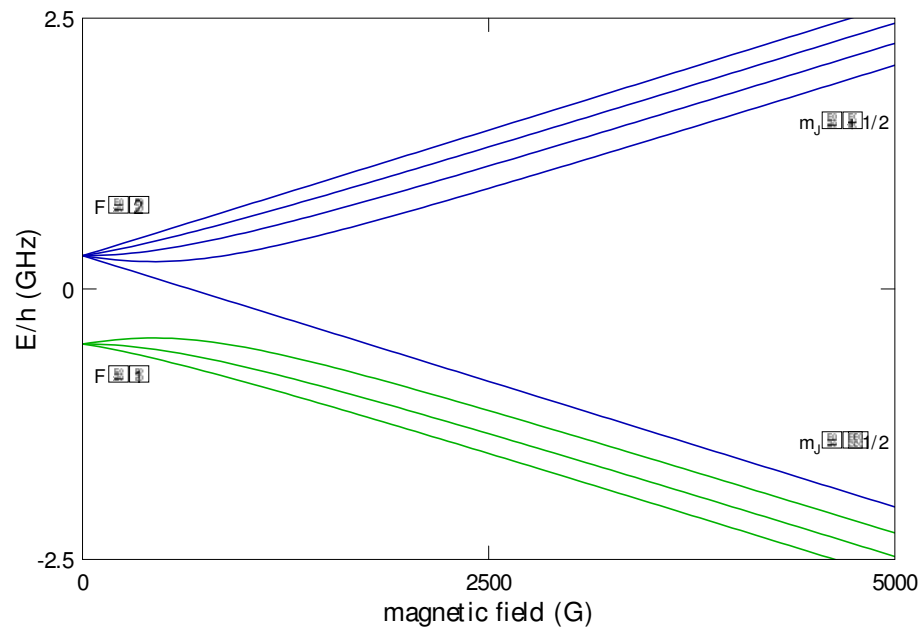


Figure 5: Rubidium 87 $5^2P_{1/2}$ (D₁ excited) level hyperfine structure in an external magnetic field. The levels are grouped according to the value of F in the low-field (Zeeman) regime and m_j in the strong-field (hyperfine Paschen-Back) regime.

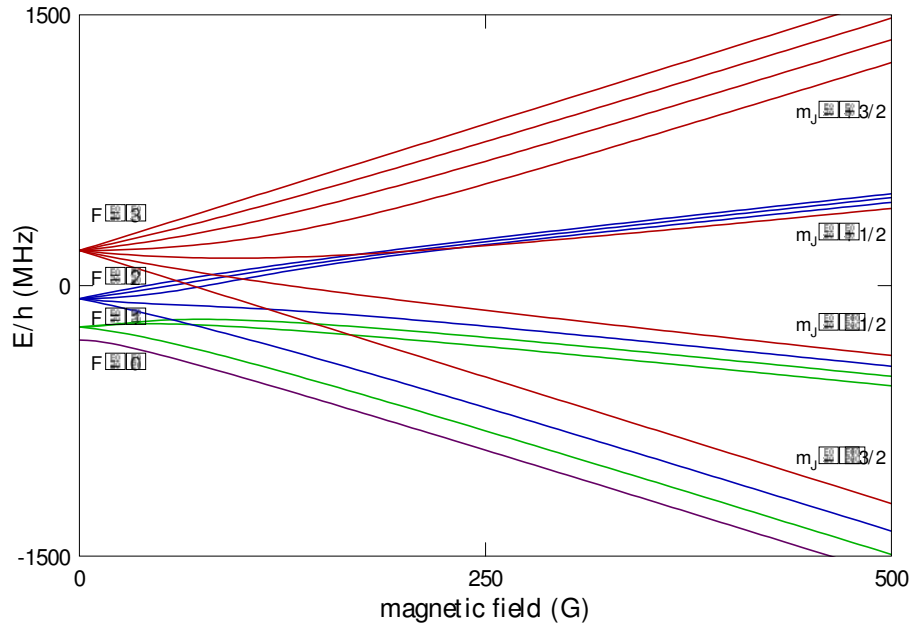


Figure 6: Rubidium 87 $5^2P_{3/2}$ (D_2 excited) level hyperfine structure in an external magnetic field. The levels are grouped according to the value of F in the low-field (Zeeman) regime and m_j in the strong-field (hyperfine Paschen-Back) regime.

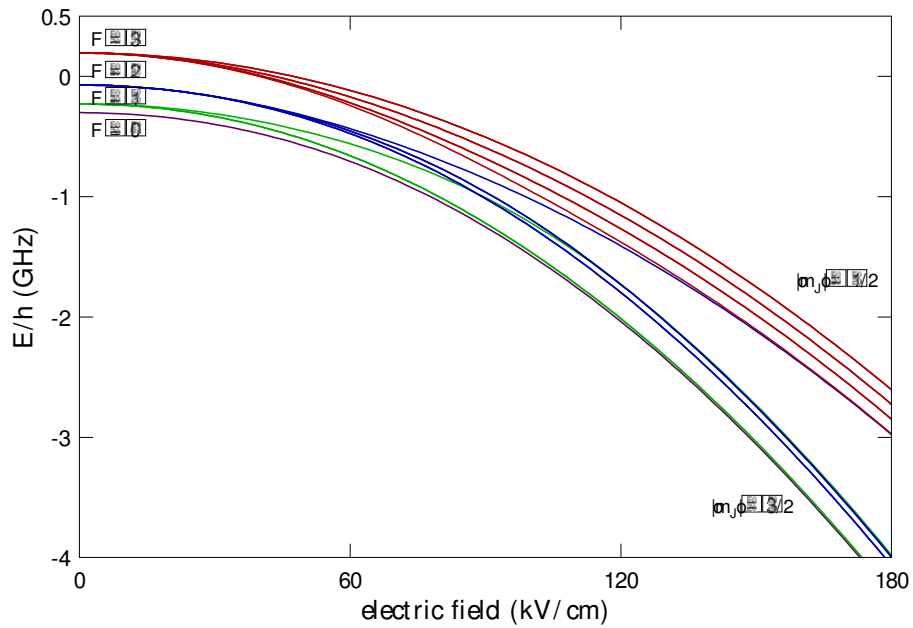


Figure 7: Rubidium 87 $5^2P_{3/2}$ (D_2 excited) level hyperfine structure in a constant, external electric field. The levels are grouped according to the value of F in the low-field (Zeeman) regime and $|m_j|$ in the strong-field ("electric" hyperfine Paschen-Back) regime. Levels with the same values of F and $|m_j|$ (for a weak field) are degenerate.

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